

MUSAWANDILE SHAMASE

Electronics & Computer Engineering Graduate (BEngTech)

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PROFESSIONAL SUMMARY

Electronics and Computer Engineering (BEngTech) graduate from Durban University of Technology with a strong technical foundation in embedded systems, electronic circuit design, control systems, and software development. Hands-on experience designing and prototyping engineering projects using microcontrollers, sensors, and low-level embedded programming, from initial concept through to working hardware. Currently completing a PLC Programming Certification to build specialized competency in industrial automation. Seeking a junior engineering, industrial automation, or graduate development role where hardware design, programming, and problem-solving skills can add immediate value.

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Arduino C++, HTML & CSS, Ladder Logic (In Progress)

Industrial Automation: PLC Architecture, Scan Cycles, Digital & Analog I/O Configuration, Ladder Logic, Timers & Counters, Automation Fundamentals

Engineering & Hardware: Embedded Systems, Circuit Design & Troubleshooting, PCB Assembly (SMT & Through-Hole), Soldering, Digital & Analog Electronics, Control Systems

Software & Tools: MATLAB, Arduino IDE, Linux, Windows, macOS, Microsoft Office Suite

Instruments & Testing: Oscilloscope, Digital Multimeter, Calibration & Testing Equipment

ENGINEERING PROJECTS

Smart Home Irrigation System

Arduino, C++, Soil Sensors, Relay Modules

- Designed and prototyped an automated irrigation control system that monitors soil moisture levels in real time to regulate water pump actuation.
- Programmed microcontrollers in C++ to optimize irrigation intervals, reducing water waste and improving overall system efficiency.

Cable Fault Detection System

Embedded Systems, Signal Processing, Sensors

- Engineered a diagnostic system to detect and pinpoint fault locations in electrical and signal cabling, reducing operational downtime.
- Integrated embedded sensor interfaces and signal processing logic to deliver precise fault detection and reporting.

Car Safety Interlock System

Microcontrollers, Embedded C, Safety Logic

- Developed an embedded safety interlock system that inhibits vehicle startup until safety checks, such as seatbelt fastening, are validated.
- Configured microcontroller hardware interrupts and firmware logic in Embedded C for real-time safety condition verification.

PROFESSIONAL DEVELOPMENT & CERTIFICATIONS

PLC Programming Certification — PLCDOJO

In Progress

- Key focus areas: PLC Architecture, Scan Cycles, Digital/Analog I/O, Ladder Logic, Timers & Counters, Industrial Automation.

WORK EXPERIENCE

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April 2026 – Present

MyChina, Boksburg, South Africa

- Maintain product displays, stock replenishment, and inventory accuracy in a high-volume retail environment.

- Demonstrate operational discipline, attention to detail, and effective team coordination to consistently achieve daily targets.

Educational Assistant

June 2025 – November 2025

KwaBulawayo High School (Department of Basic Education), South Africa

- Supported educators with lesson preparation, classroom administration, and academic tracking.
- Delivered structured academic guidance and tutoring to high school learners in core STEM subjects.

Volunteer Programming Tutor

March 2022 – August 2024

Dataholics Student Organisation, Durban, South Africa

- Tutored undergraduate students in computer programming fundamentals, logic design, and software debugging.
- Conducted practical coding workshops and test preparation sessions that improved academic outcomes.

EDUCATION

Bachelor of Engineering Technology (BEngTech) — Electronics & Computer Engineering *Graduated 2025*

Durban University of Technology (DUT), Durban, South Africa

ADDITIONAL INFORMATION

- Driver's Licence: Code 10 (C1)
- Relocation: Open and available to relocate across South Africa
- Work Eligibility: South African Citizen — eligible to work immediately
- References available on request